Quantum Brain Ultrasound Explained

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1. The medical problem

A stroke patient needs a brain scan before treatment, and today that scan usually waits for a hospital. A **clot** is treated with a clot-busting drug; a **bleed** is made worse by that same drug, so doctors must see which one it is first.

CT and MRI scanners are large and stay in the hospital, so many patients wait an hour or more. Every minute of delay kills brain tissue.

The paper (Guasch et al. 2020) proposes a helmet of 1,024 small ultrasound speakers and microphones (transducers) that could ride in an ambulance. Ultrasound normally can't see through the skull; the paper's computational method gets around that.

2. What the paper's algorithm does

The algorithm, full-waveform inversion, builds an image of the head by guessing and checking against real recordings. Sound travels at different speeds through different tissue: about 2,800 m/s in skull, 1,546 m/s in brain and 1,578 m/s in blood. A map of sound speed inside the head is therefore an image of the head, and a bleed shows up as a slightly faster patch.

1. **Guess** the inside of the head: the sound speed at every point.
2. **Simulate** how sound from each speaker would travel through that guessed head, and what each microphone would hear.
3. **Compare** the simulated recordings with the real ones.
4. **Adjust** the guess to shrink the mismatch, then repeat until they match.

Step 2 is the expensive one, because it runs for 1,024 speakers in every round. On a cluster of 128 computers the paper's run took 32 hours. That simulation step is where quantum computing might help.

3. How a computer simulates sound

The computer chops the head into a 3D grid of tiny cubes, 0.5 mm on a side, like a 3D pixel image. Each cube stores the current sound pressure.

To move time forward one tick, each cube updates from its neighbours: if they push harder, its pressure rises. Skull cubes pass the push along faster than brain cubes.

A head is about 22 cm across, so the grid is 440 × 440 × 440, about **85 million cubes**. Each is updated thousands of times per speaker, which is why the paper needed a supercomputer.

Our 1D version uses the same rule on a single line of 7 cubes from ear to ear: scalp, skull, skull, brain, skull, skull, scalp. In 3D each cube has 6 neighbours (left/right, front/back, up/down); in 1D it has 2. The physics and the update rule do not change. The line is the smallest version that still contains a skull and a bleed, so it works as a unit test for the method.

4. Quantum computing in four ideas

A quantum computer can hold enormous numbers of values at once, but it only lets you sample them one click at a time.

1. **A qubit.** A normal bit is 0 or 1. A qubit can be in a blend of both, described by two numbers saying how much of each.
2. **Qubits multiply.** 2 qubits cover 4 combinations (00, 01, 10, 11), each with its own number; 3 cover 8; *n* qubits cover 2^n. So **30 qubits carry about a billion numbers**, where a normal computer needs a billion memory slots.
3. **You can't read the numbers out.** Measuring collapses the blend to one random combination, like `0110`. Combinations with bigger numbers come up more often. To learn the numbers you rerun the whole thing thousands of times and count; each run is a **shot**.
4. **Gates move the numbers by fixed rules.** Operations called gates shift and mix the numbers while keeping one total fixed: the squares of all the numbers always add to 1.

5. Why sound waves fit a quantum computer

Sound conserves energy the way a quantum computer conserves its total, so the wave equation can be rewritten in the form a quantum computer runs. Energy sloshes from cube to cube, but the total stays fixed (ignoring absorption). Costa, Jordan & Ostrander

(2019) showed the rewrite. After it:

- Each of the 2^n numbers stands for **the wave energy in one cube**.
- Gates move energy between neighbouring cubes, just as real sound does.
- A measurement lands in one cube. **How often it lands on a microphone's cube tells you how much sound reached that microphone.**

Our prototype did exactly this. The circuit's rate of landing on the microphone cube matched the classical simulation.

6. Two ways to store the head, and the catch in each

The storage that gives quantum its advantage can't run on today's chips, and the storage that runs today gives up the advantage.

A. Compact storage. Number the cubes and store the wave using idea 2: 85 million cubes, plus the in-between edges the method needs, fit in about **29 qubits**. A supercomputer needs gigabytes of memory for the same thing.

The catch: moving energy between neighbours is complicated when cubes are stored by number. Even our 7-cube test needed about 1,100 two-qubit gates for good accuracy. Today's IBM chips make an error roughly every few hundred two-qubit gates, so the result comes out as noise.

B. One qubit per cube. Each cube gets its own qubit, and the wave is a single "1" hopping along the line of qubits. Neighbour moves become simple gates between neighbouring qubits: about 240 gates for the same accuracy, roughly 5× cheaper.

The catch: 85 million cubes would need 85+ million qubits, and IBM's largest chips have about 156. The memory advantage is gone, and a normal computer can simulate this version easily.

7. How 1D becomes 3D

Going from our line to the paper's head multiplies memory by millions, but the quantum version saves memory, not time steps.

	Our 1D test	The paper's 3D head
Cubes	7	~85 million

	Our 1D test	The paper's 3D head
Neighbours per cube	2	6
Qubits, compact storage	4	~29
Qubits, one per cube	15	~300 million (impossible)
Steps needed	~8	thousands, to cross ~440 cubes
Two-qubit gates	hundreds to ~1,000	estimated millions or more
What we can do	run it on real IBM hardware	calculate what it would take

The step count barely changes because sound moves about one cube per tick on any computer. The quantum circuit needs about as many steps as the normal simulation. The 3D gate count is a rough estimate; needing future error-corrected machines is the safe conclusion.

8. The big wall: getting the answer out

Reading results out of a quantum computer is the main reason it doesn't solve the paper's problem, even in theory. Every round, the algorithm needs the full recording from every microphone, for every speaker, at every moment: roughly 1,024 × 1,024 × 1,000 ≈ **a billion numbers**.

- A GPU simulation hands back all of them at once.
- A quantum computer gives one click per run. One number to 1% accuracy takes about 10,000 runs, and the algorithm needs a billion numbers per round.

Quantum could help with questions that have a **single-number answer**, such as "how badly does this guessed head mismatch the recording overall?" or "is there a bleed: yes or no?" So the design is hybrid:

- **Quantum:** simulate the wave and estimate one mismatch number.
- **Classical:** decide how to adjust the guess, build the image, make the diagnosis.

9. What the team can honestly claim

Every claim below is something the team built or measured, which makes the argument hard for judges to knock down.

1. We turned the paper's sound simulation into a quantum circuit, and it matches the classical simulation.
2. We ran a small version, a line through the skull, on a real IBM quantum computer and measured how much noise damages it and how much error-reduction tricks recover. *(Still to do.)*
3. We calculated what the full 3D head would need: about 29 qubits of memory, but millions of error-free operations and an impossible number of shots for full readout.
4. Quantum fits the physics, but today it loses to GPUs on gate count, noise and readout, and we can say what would have to improve.

Sources

- Guasch et al. 2020, Full-waveform inversion imaging of the human brain, npj Digital Medicine
- Costa, Jordan & Ostrander 2019, Quantum algorithm for simulating the wave equation
- A quantum computing concept for 1-D elastic wave simulation (Qiskit, IBM Brisbane)